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Doping level dependence of low-frequency spin dynamics in $La_{2-x}Ba_xCuO_4$ studied by Cu-NQR

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Abstract

In order to study the low-frequency spin dynamics in the doped CuO_2 square lattice in $La_{2-x}Ba_xCuO_4$, the nuclear spin-lattice relaxation time T_1 of ^{63}Cu was measured up to about 900 K. Overall temperature dependence of $(T_1T)^{-1}$ cannot be represented by a single Curie–Weiss (CW) law. At temperatures higher than a crossover temperature T^*, $(T_1T)^{-1}$ obeys the high-T CW law, and is suppressed below T^*, following another low-T CW law. The x-dependence of the CW parameters and T^*, which is very similar to those for $La_{2-x}Sr_xCuO_4$, was discussed referring to the current theoretical models. © 2000 Elsevier Science B.V. All rights reserved.

Keywords: NQR; High-T_c cuprates; Spin dynamics; Spin gap; Metal–insulator transition

Doped CuO_2 square lattice of high-T_c superconductors have been attracting much interest in the studies of strongly correlated electron system. The Cu nuclear spin-lattice relaxation rate, T_1^{-1}, generally provides us with information on the low-frequency spin dynamics of the CuO_2 plane. The doped La_2CuO_4 system is a prototype for high-T_c superconductors to study the doping dependence of the spin dynamics. Especially in $La_{2-x}Sr_xCuO_4$, the doping dependence of $(T_1T)^{-1}$ has been studied by many groups, so far [1–5]. The following have been found: (1) T-dependence of $(T_1T)^{-1}$ shows a crossover from high-temperature Curie–Weiss (HT-CW) law, $C_H/(T+\theta_H)$, to low-temperature Curie–Weiss (LT-CW) one, $C_L/(T+\theta_L)$. (2) The sign of θ_H changes around $x=0.13$, which is close to optimal doping, from a negative value to a positive one with increasing doping. (3) Below the crossover temperature T^*, there is the unconventional suppression of $(T_1T)^{-1}$, which may be a hallmark of spin gap in this system. (4) T^* increases with decreasing doping.

In this paper, we present x-dependence of $(T_1T)^{-1}$ in $La_{2-x}Ba_xCuO_4$. The measurements were carried out up to about 900 K for the samples with $x=0.04$, 0.08, 0.125 and 0.15, using phase coherent spin–echo method.

Fig. 1 shows the T-dependence of T_1T of planar ^{63}Cu. Two different linear dependences of T_1T below and above T^* indicate that $(T_1T)^{-1}$ in this system also follows two CW laws as described above in (1), and a crossover occurs in the nature of the low-energy spin dynamics. The CW parameters θ_H, θ_L, C_H, C_L and T^* are shown in Fig. 2, together with those in $La_{2-x}Sr_xCuO_4$ [1,4,5]. The doping dependence of these parameters is very similar to those for $La_{2-x}Sr_xCuO_4$. Thus, these features concerning low-energy spin dynamics described above in (1)–(4) are common in both the compounds. We discuss these properties below referring to the current theoretical model.

Since $(T_1T)^{-1}$ is unusually suppressed below T^*, HT-CW behavior must be considered primarily rather than the LT-CW one. The CW T-dependence of $(T_1T)^{-1}$ has been well interpreted by 2D-SCR theory [6,7] or the phenomenological spin fluctuation model [8,9]. According to SCR theory, $(T_1T)^{-1}$ follows Curie law ($\theta=0$) at a magnetic order–disorder transition at $T=0$ K, and both sides with order and disorder,

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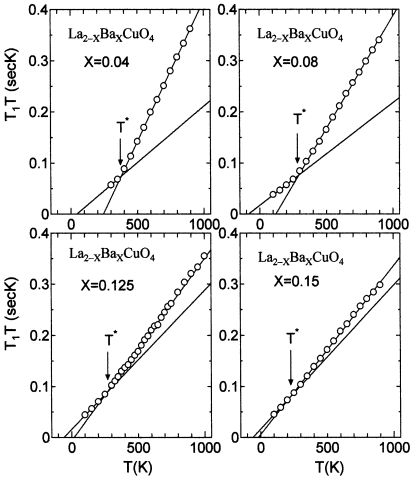


Fig. 1. Temperature dependence of T_1T for ^{63}Cu in $La_{2-x}Ba_xCuO_4$.

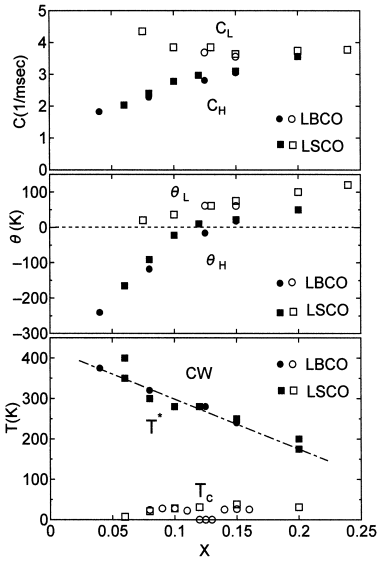


Fig. 2. X-dependence of CW parameters, $\theta_H, \theta_L, C_H, C_L$ and the crossover temperature T^* in $La_{2-x}Ba_xCuO_4$ and $La_{2-x}Sr_xCuO_4$ [4,5].

$(T_1T)^{-1}$ follows CW law with positive and negative θ, respectively, at higher temperature than T^*_{SCR}. The experimental finding of the sign change may suggest that the magnetic boundary exists near $x = 0.13$ far from the actual metal–insulator (MI) transition point. SCR theory also predicts a suppression of $(T_1T)^{-1}$ below T^*_{SCR} in such a way that T^*_{SCR} decreases to zero towards the magnetic boundary. This is not consistent with the x-dependence of T^* as far as the actual MI transition is assumed to correspond to the magnetic boundary.

The x-dependence of T^* is also hard to explain in the framework of doped 2D quantum Heisenberg AF model (2D-QHAF) [10,11]. This model assigns the ground state of the underdoped metallic phase to quantum disordered regime with a spin excitation gap. The excitation gap is expected to decrease with decreasing doping towards quantum critical regime at MI transition. The x-dependence of the excitation gap is opposite to that of T^* as long as the suppression of $(T_1T)^{-1}$ is attributed to this gap.

If we assume the suppression of $(T_1T)^{-1}$ as a sign of the spin gap observed widely as a peak of $(T_1T)^{-1}$ in many underdoped high-T_c superconductors, the doping dependence of T^* is qualitatively consistent with that of T_{RVB}, the onset temperature of spinon singlet formation, in the mean-field t–J model [12]. However, it is required to explain why $(T_1T)^{-1}$ does not show such a prominent peak as expected at T_{RVB} in the t–J model.

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